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M-011 MISCELLANEOUS HOISTS TROLLEYS

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M 011 MISCELLANEOUS HOISTS TROLLEYS

1 SCOPE OF WORK

The Scope of Work shall be in accordance with Part 4, Subsection 1.2 Summary of works.

The Bidder shall furnish, install and commission a complete system of miscellaneous hoists and trolleys to perform maintenance and inspection of all equipment furnished by him including but not limited to those as listed below:

- Water Treatment Plant
- Chlorination Plant
- Steam turbine ancillary equipment
- Feedwater / Condensate pumps
- Air compressors
- Emergency Diesel Generator
- Other Installations

All fixed trolley beams and moving steel trolleys shall be provided by the Bidder.

2 CODES AND STANDARDS

The Bidder shall comply with the requirements of Part 4 Subsection 2.5.

The codes and standards of the following organizations or other equivalent international standards are specifically applicable to the design, manufacture and testing of the equipment in this specification.

ASTM	American Society of Testing and Materials
ANSI.	American National Standards Institute, Inc.
CMAA	Crane Manufacturers Association of America
IEC	Electro Technical Commission
NEC	National Electrical Code
NEMA	National Electrical Manufacturers Association

3 OPERATING AND DESIGN CONDITIONS

Hoists and/or trolleys will be used for maintenance and equipment repair.

Bidder shall essentially use the manufacturer's standard product, except as specified otherwise to meet the operating conditions specified. If the specified lifts are not available, the Bidder shall bid the nearest standard lift.

Design and construction throughout shall be based on ample safety factors so as to guard against injurious strains and distortions.

All hoist parts shall be designed with a minimum safety factor of 5. All parts shall be readily accessible for inspection, adjustment and repair.

4 DETAILED REQUIREMENTS

4.1 Trolleys

Each trolley shall contain all mechanisms for operating the hooks and for moving the trolley on the rails.

Trolley wheels shall be of forged steel or hardened steel. Cast iron or cast steel shall not be used.

4.2 Hoists

Wire rope : Non-spinning wire rope shall be used on all hoists where the load is suspended by a single rope.

Hooks : Hooks shall be of forged steel supported on a ball or roller bearing. They shall rotate freely within this bearing.

4.3 Gearing

All gears shall be forged, rolled or welded steel with cut teeth. All gearing shall be 20° full depth spur, helical or herringbone type. All gearing shall be enclosed in dust tight housing.

All gearing shall operate in grease lubricant or oil as appropriated.

4.4 Bearings

All bearings, except those subject to a small amount of movement and as approved, shall be of the antifriction type with the balls or rollers retained in a cage.

Bearings shall be grease lubricated. Bearings shall be sealed to prevent leakage and all bearings shall be protected against entry of dust.

4.5 Lubrication

Bearings shall be grease-tight. The equipment shall be arranged so that all points of lubrication are easily and safely accessible.

4.6 Guards

Shields or enclosures shall be provided over any part of the equipment which needs protection from water or dust.

Exposed moving parts, such as gears, projecting keys, set screws etc. that might constitute a hazard under normal operating conditions and as required by safety codes, shall be equipped with substantial guards. Guards shall be securely fastened.

4.7 Painting

Before shipment all parts of the equipment shall be thoroughly cleaned of all mill scale, rust, grease and other foreign matter and shall be painted in accordance with the requirement as mentioned in S-420 and S-430 except otherwise Specified.

Primer paint shall be manufacturer's standard, suitable for the environmental conditions of the project site.

4.8 Protective Coating

Before shipment all exposed machined surfaces shall be coated with a suitable grease or antirust compound.

4.9 Tools

Any special tool shall be in accordance with Part 4, Subsection 2.11.

5 PERFORMANCE TESTING GUARANTEES

5.1 Shop Tests

The following tests shall be performed on each hoists and trolleys by the Bidder under the direction of the Owner.

The hoist shall be operated to raise, lower and hold in any position a test load of 125% of the rated capacity.

All safety devices shall be tested.

5.2 Site Tests

5.2.1 Running in Period Test

The hoists and trolleys shall be functionally tested by the Bidder. The Bidder shall make any adjustment or modification required to the hoist and trolley system for compliance with the specified service at no extra cost to the Owner.

6 QUALITY ASSURANCE/QUALITY CONTROL (QA/QC)

Quality assurance/control shall conform to the requirements of Part 4. Subsection 2.7.

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